

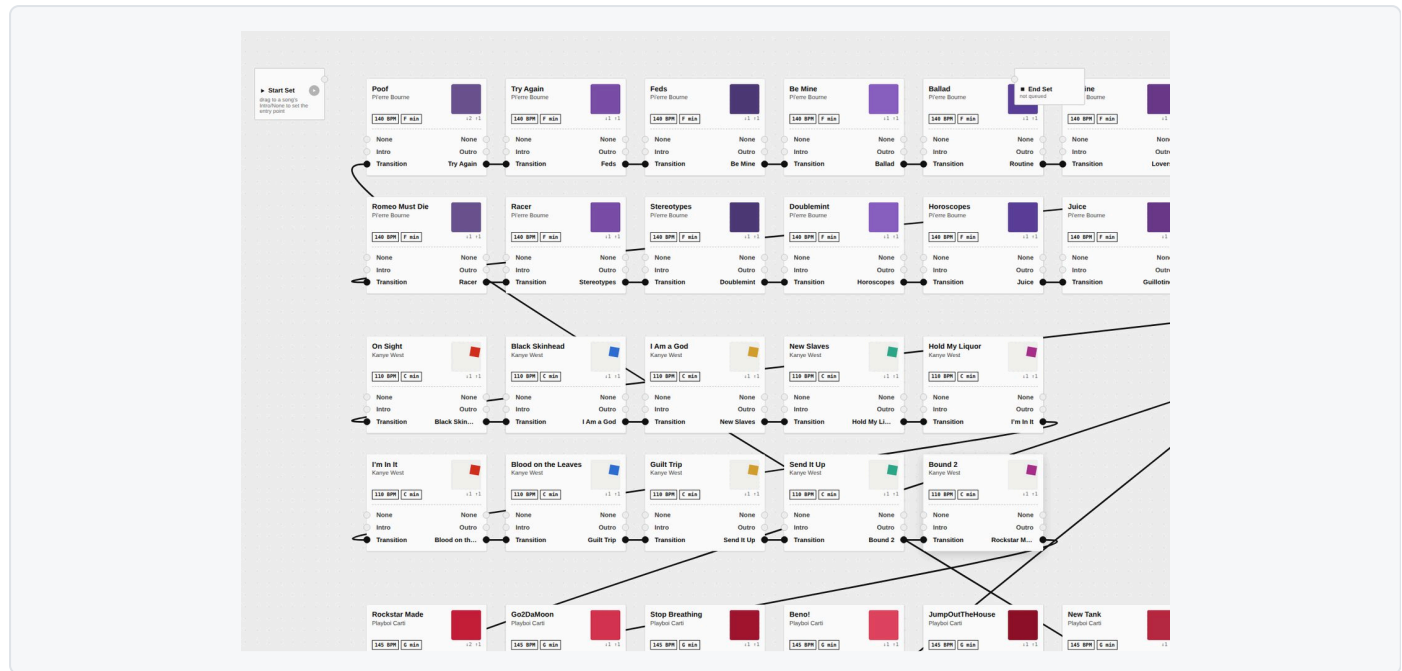
What professional node editors do that Eskimo doesn't

Ten node-based interfaces, from utilitarian workhorses to colour-forward modern tools — and the specific, borrowable decisions that make them read as professional software.

01	Eskimo today: an honest diagnosis	09	ComfyUI
02	The spectrum	10	n8n
03	Nuke	11	Unreal Blueprints
04	Houdini	12	Bitwig — The Grid
05	Blender	13	What they all agree on
06	TouchDesigner	14	Recommendations for Eskimo
07	Audulus	15	Proposed node anatomy
08	cables.gl	16	Sources

Eskimo today: an honest diagnosis

Before borrowing anything, it's worth being precise about what actually looks wrong. Most of it is not styling — it's information density.



Eskimo Studio, current build. A representative crop of the example graph at 1:1. Fifty cards, one size, one colour, six socket rows each.

1 The socket matrix is always fully expanded.

Every card prints six rows — None / Intro / Transition on the left, None / Outro / Transition on the right — whether or not they carry anything. Across fifty nodes that is three hundred rows of mostly-repeated words. It is the single largest source of visual noise, and no other tool in this study does it.

2 The word "None" appears twice on every card.

Once per side. It labels the absence of a thing. Roughly a hundred instances of the same non-information across the canvas.

3 Every wire looks identical.

Uniform black, uniform weight, no directionality, no type. A transition, a cold cut and an outro handoff are visually indistinguishable — even though the difference between them is the entire point of the app.

4 Wires route straight through other cards.

Long diagonals cross unrelated nodes, so the card boundary stops reading as a boundary. Every tool here solves this with curvature, orthogonal routing, or both.

5 There is exactly one node size.

A four-input hub and a leaf node occupy identical space. Importance is not expressible, and zooming out doesn't simplify anything — it just makes the same text unreadable.

6 Colour is spent on decoration, and there's no figure/ground.

The only colour is the album swatch — which tells you which record the song is from, something the title beside it already said. Meanwhile white cards on light grey with a 1px shadow leave almost no value gap, so the graph reads as a flat sheet of text rather than as objects in space.

Worth saying plainly: the interaction model is fine — toggling sockets, the variant picker, drag-to-connect, autoarrange, multi-select. Several tools in this study do less. This is a presentation problem, not an architecture one.

The spectrum

Ten tools, placed by how much visual personality they spend. None of these positions is "better" — they're different answers to how much the interface should assert itself over the work.



BORING BUT RELIABLE

Nuke
Foundry · compositing

Houdini
SideFX · procedural 3D

Blender
shader & geometry nodes

SLEEK AND MODERN

TouchDesigner
Derivative · realtime

Audulus
modular synthesis

cables.gl
browser-native creative coding

COLOURFUL AND INTUITIVE

ComfyUI
diffusion pipelines

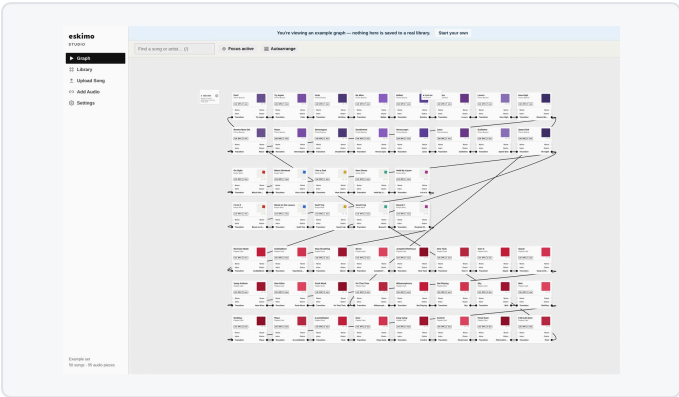
n8n
workflow automation

Unreal Blueprints
visual scripting

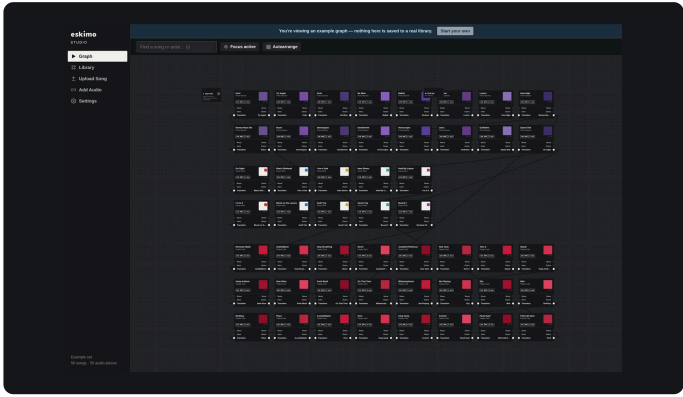
Bitwig — The Grid
modular audio

Where Eskimo sits, and where it could sit

Eskimo is currently at the far left by default rather than by decision — it is restrained, but restraint isn't the same as reliability, and the restraint isn't buying density. The most useful target is around **n8n** and **Bitwig**: a light, calm canvas that still uses colour to carry meaning, with one strong idea (signal type) driving the palette.



Eskimo, light. The default and the one that matters most.



Eskimo, dark. Already implemented. Note how much better the cards separate from the canvas here — the value gap does the work.

03 Nuke

Foundry · film compositing · the DAG

BORING BUT RELIABLE

Nuke's node graph is close to the minimum viable node editor, and it has survived three decades of professional use in exactly that form. Nodes are small coloured bars carrying a name. Flow reads top-to-bottom. There is almost no chrome.

RECREATION — NUKE NODE CONVENTIONS (NOT A SCREENSHOT)



THE DESIGN SYSTEM

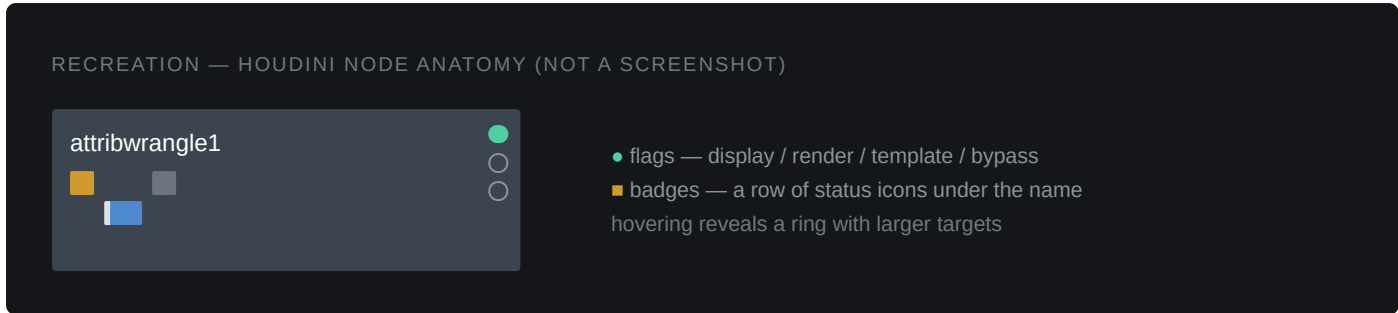
- **Node colour encodes class** — reads, merges, writes and filters each get a family hue, and users can recolour any node to mark up their own script.
- **The node is the label.** No subtitle, no metadata, no icon. Size is driven by the name.
- **Postage stamps** — optional thumbnails on the node showing what it passes downstream. Critically, they can be set to a static frame or switched off entirely, because on a large script they cost real performance.

STEAL THIS FOR ESKIMO

The postage stamp, and its off switch. A song node showing a waveform with its cue points marked is the direct analogue, and it would make the graph genuinely informative rather than decorative. Nuke's hard-won lesson is that the thumbnail must be optional — Eskimo already renders fifty album swatches with dominant-colour extraction, so the same "turn it off on big graphs" affordance belongs there too.

Colour by class, not by decoration. Nuke spends its entire colour budget on "what kind of node is this". Eskimo currently spends its budget on album art.

Houdini packs more state onto a node than anything else in this study, and does it without the node becoming a mess. The trick is that state is expressed as small, fixed-position marks rather than words.



UX FEATURES WORTH NOTING

FEATURE	WHAT IT SOLVES
Flags	Persistent per-node state (which node is the output, what's bypassed) as always-visible toggles on the node body, not buried in a panel.
Badges	A row of small icons beneath the name carrying status — errors, warnings, comments, time dependency. Glanceable across a whole network.
The node ring	Hovering a node summons a contextual ring with big click targets for those flags. It exists specifically because the flags themselves become unclickable when zoomed out.
Network boxes	Labelled, coloured, resizable containers that group nodes and move them together.

STEAL THIS FOR ESKIMO

The badge row. Eskimo has at least four things it currently either hides or spells out in words: has produced audio, verified splice, local-only vs synced, and contributed-by. Four small marks under the title would replace a lot of text and make the whole library state visible at a glance.

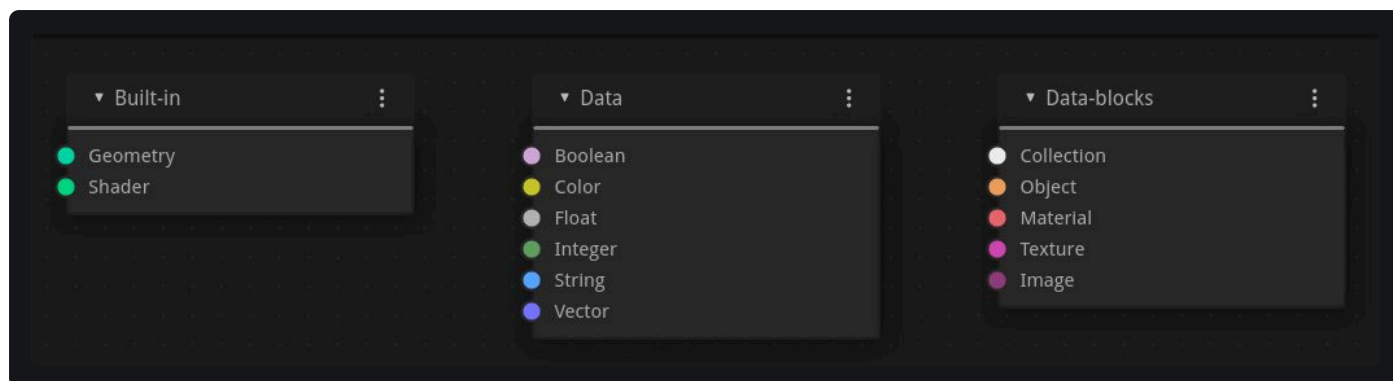
The hover ring is the answer to Eskimo's tiny sockets. The socket dots are ~9px and sit outside the card; at any zoom below 100% they're near-impossible to hit. Houdini hit this exact wall and solved it by escalating the target on hover rather than making the resting state bigger.

05 Blender

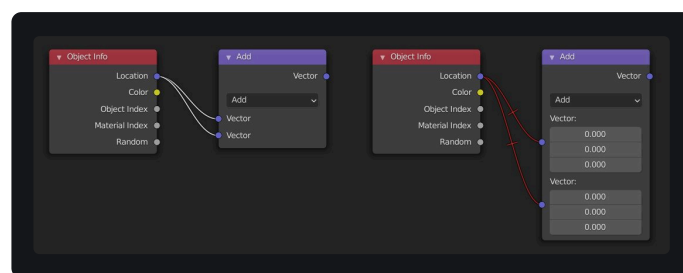
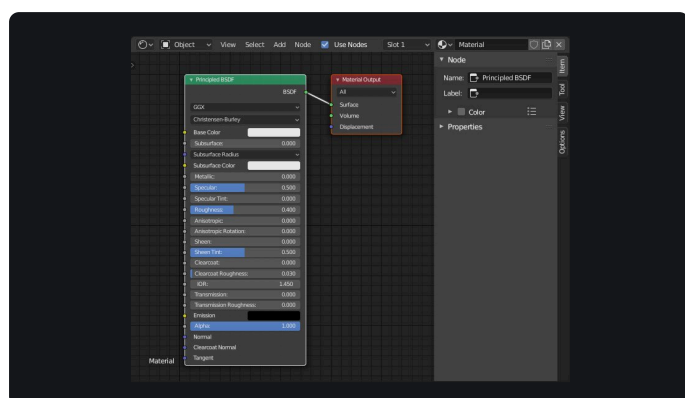
BORING BUT RELIABLE

Shader, compositor and geometry node editors

Blender's node editors are the best mainstream example of a *rigorously applied type system expressed as colour*. It is not decorative; the colour is the documentation.



The socket colour legend. Every data type gets one colour, and that colour is identical on the socket, on the wire, and in every editor. Geometry green, boolean pink, float grey, integer green, string blue, vector purple; data-blocks get their own family — object orange, material red, texture pink, image purple.



Muted links render differently — a disabled connection stays visible and legible rather than disappearing.

Header colour by category, and node width set by content.

THE DESIGN SYSTEM

- **One colour per type, everywhere.** Learn it once, apply it in every editor.
- **Collapse to a title pill.** Any node folds down to just its header — the primary density control, available per node.
- **Sockets are labelled with what they are** ("Geometry", "Color"), not with their state.

STEAL THIS FOR ESKIMO

Commit to one socket-type palette and use it on the wires too. Eskimo has exactly four connection types — None, Intro, Outro, Transition. Four colours, applied to the socket dot *and* the wire, would make the graph self-documenting and directly fix diagnosis #3.

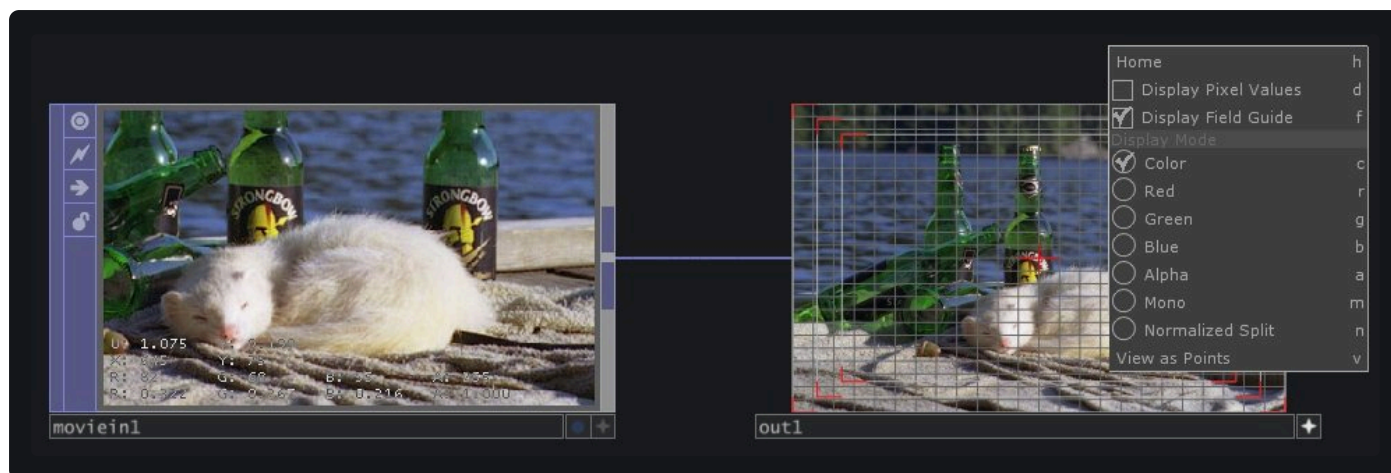
Collapse-to-pill. The cheapest possible fix for the density problem, and it's per-node so the user chooses what matters.

06 TouchDesigner

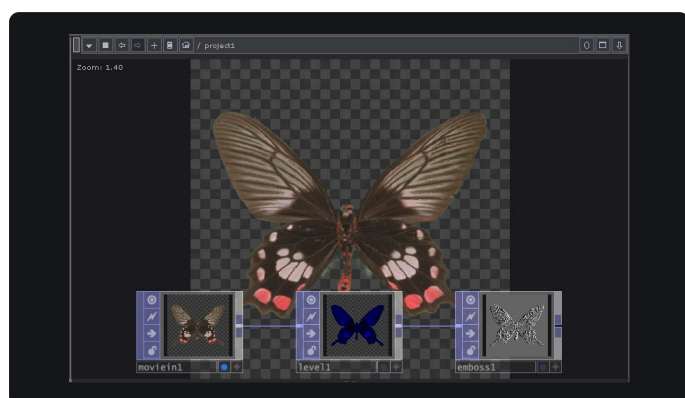
Derivative · realtime visual systems

SLEEK AND MODERN

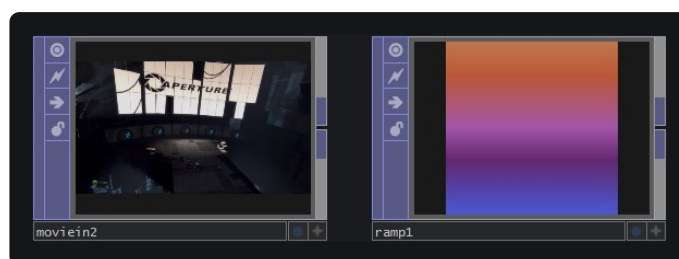
TouchDesigner's central idea is that **a node should show its own output**. The node isn't a label for a process, it's a live window onto it. On a dark canvas with restrained chrome, the content becomes the colour.



Nodes as live viewers. Each operator renders its current output in its own body, with a right-click menu controlling exactly what the viewer displays — channels, field guide, pixel values. The node name sits in a strip along the bottom.



Per-node display flags along the node's edge.



Operator family colour. TOP, CHOP, SOP, DAT, MAT and COMP each own a hue — the primary wayfinding device in a large network.

STEAL THIS FOR ESKIMO

Put the waveform on the card. This is the highest-value idea in the whole study for Eskimo specifically. A song node that shows its waveform with the outro splice point marked turns the graph from a diagram of the set into a picture of the set. The data already exists — `rmsEnvelope()` is already implemented and already used for the transition preview.

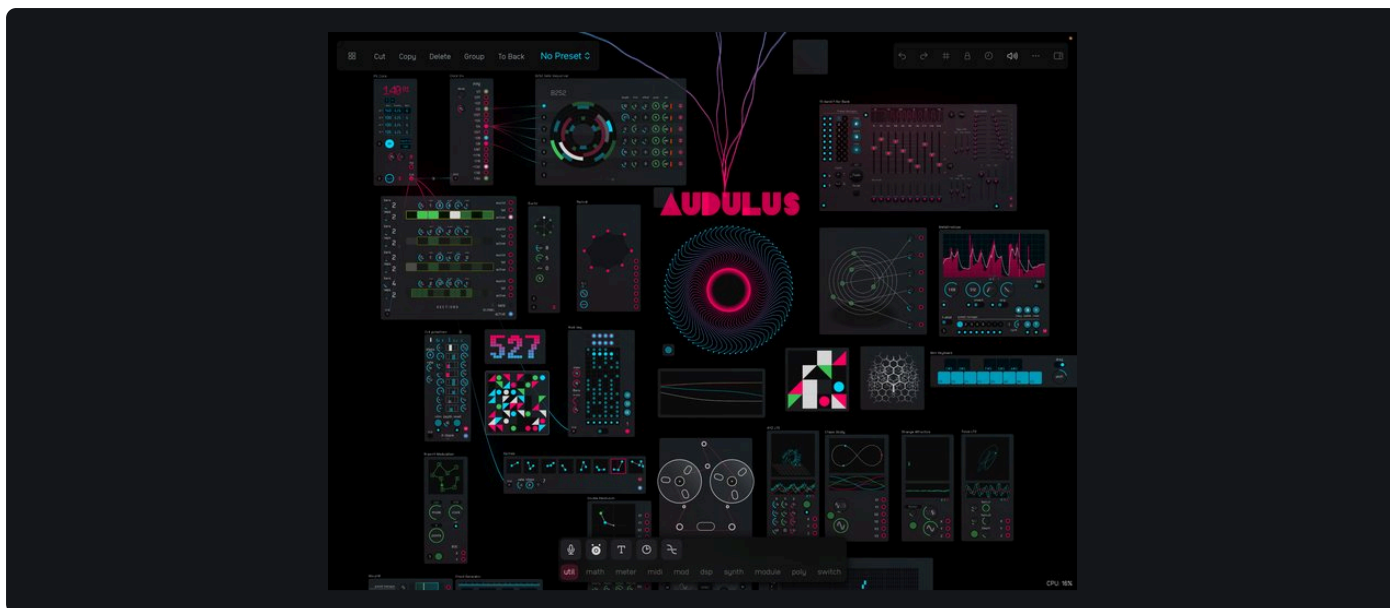
Family colour as wayfinding. Eskimo's equivalent axis is artist, BPM band, or key — any of which would give the canvas structure at a glance.

07 Audulus

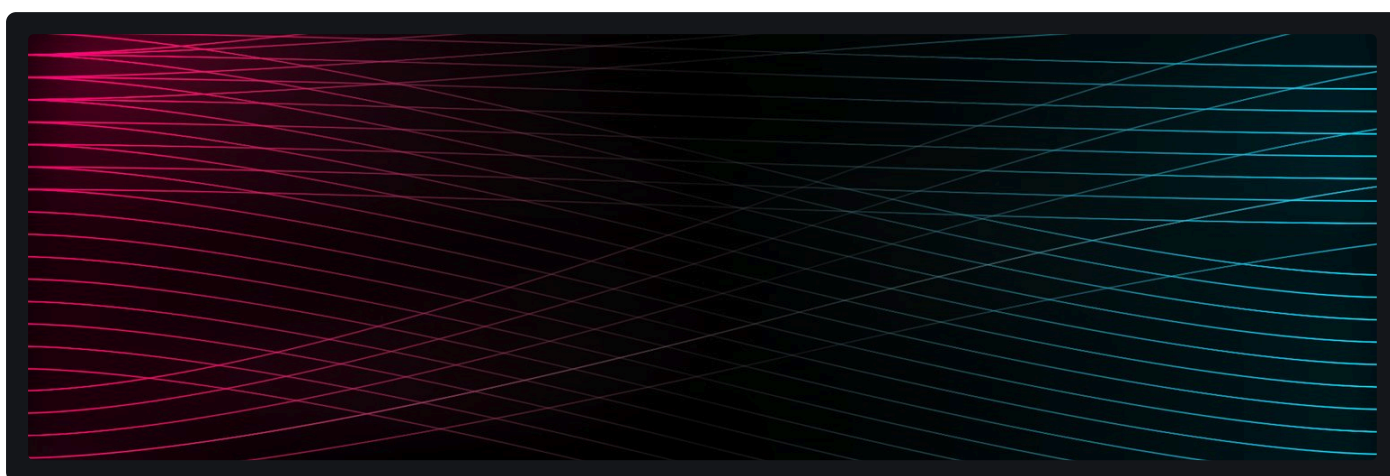
Modular synthesis · desktop and iPad

SLEEK AND MODERN

The most visually confident tool here, and the closest to "sleek" as an explicit goal. Pure black canvas, hairline neon wires, and a strict two-hue accent palette.



Audulus. Modules read as instrument panels rather than boxes of text. Note the category tab bar pinned along the bottom — util, math, meter, midi, mod, dsp, synth, module, poly, switch — so the taxonomy is permanently visible instead of hidden behind a search box.



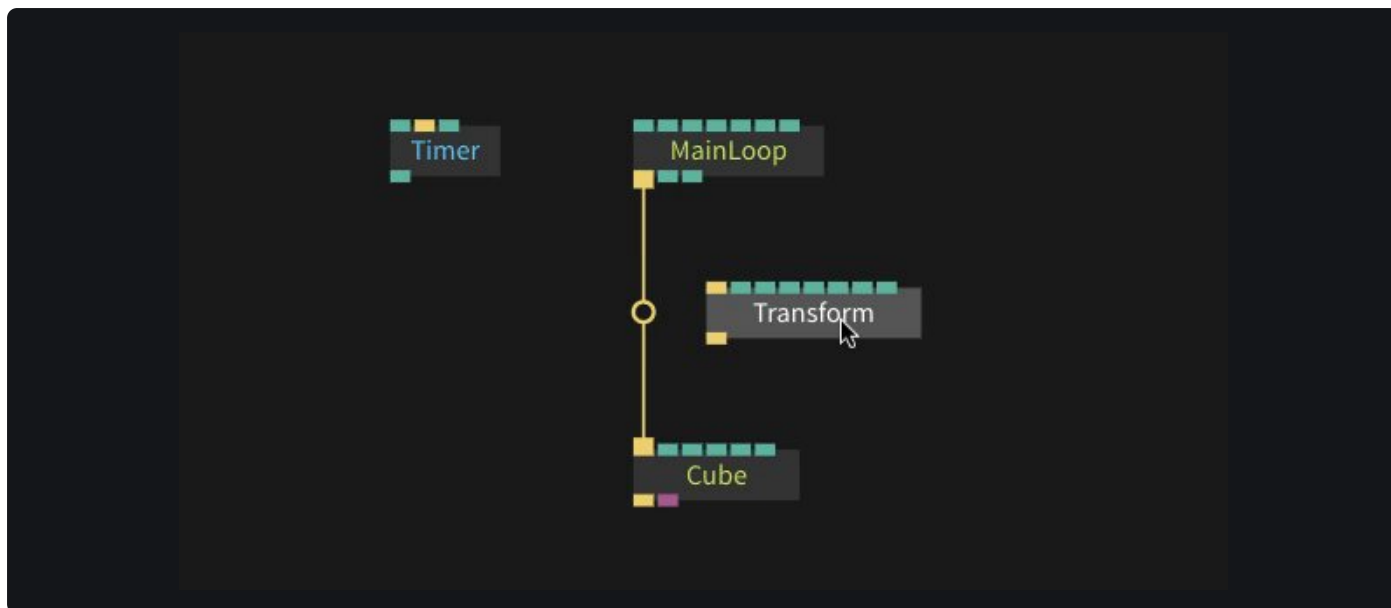
The wire aesthetic, isolated. Sub-pixel-thin strokes with a slight glow, gently curved, high chroma against near-black. Compare to Eskimo's uniform 2px black lines.

STEAL THIS FOR ESKIMO

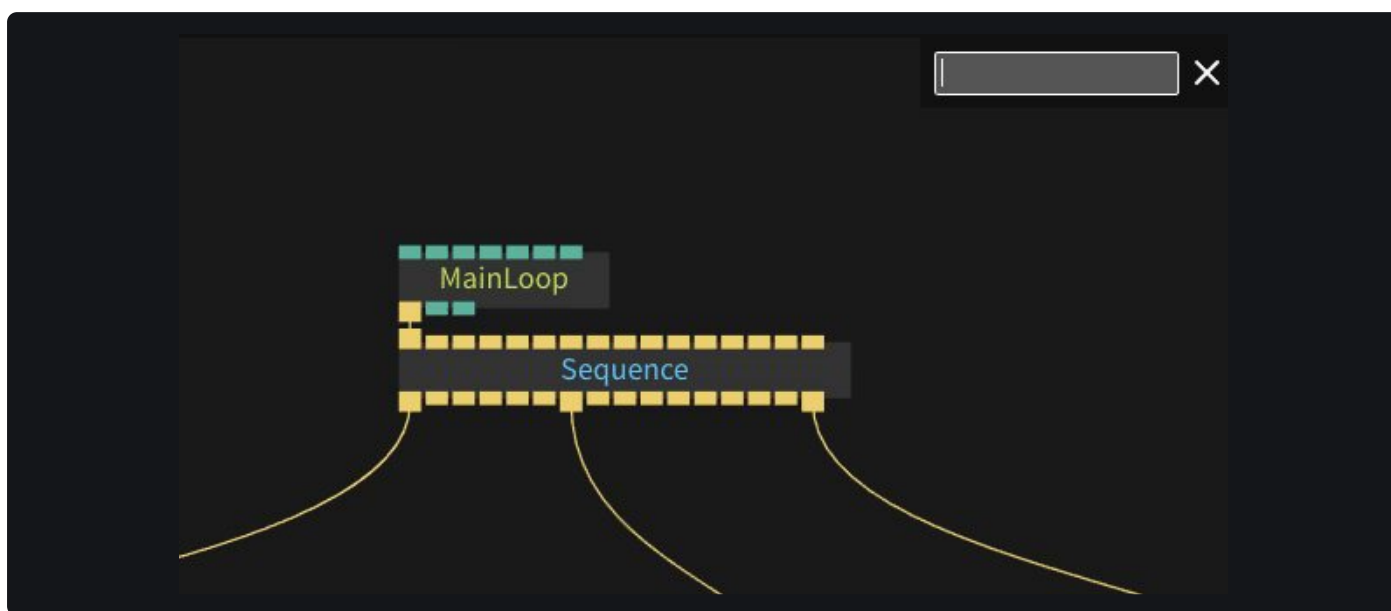
Wire craft is most of the perceived polish. Thin beats thick. Curved beats straight. A subtle glow or outer stroke lets wires cross without visually merging. This is a pure CSS/SVG change with no data model implications and would probably move Eskimo's perceived quality more than anything else on this list.

A permanent category palette. Eskimo hides "add a node" behind a right-click. A docked strip would make the vocabulary of the app visible.

The most radical density answer in the study, and the most directly applicable to Eskimo's specific problem. cables.gl has many ports per node — and spends almost no space on them.



Ports as edge strips. Each node is a plain rectangle with its name. Ports are small coloured blocks butted together along the top and bottom edges — colour carries the type, position carries the identity. A node with eight ports is barely larger than one with two.

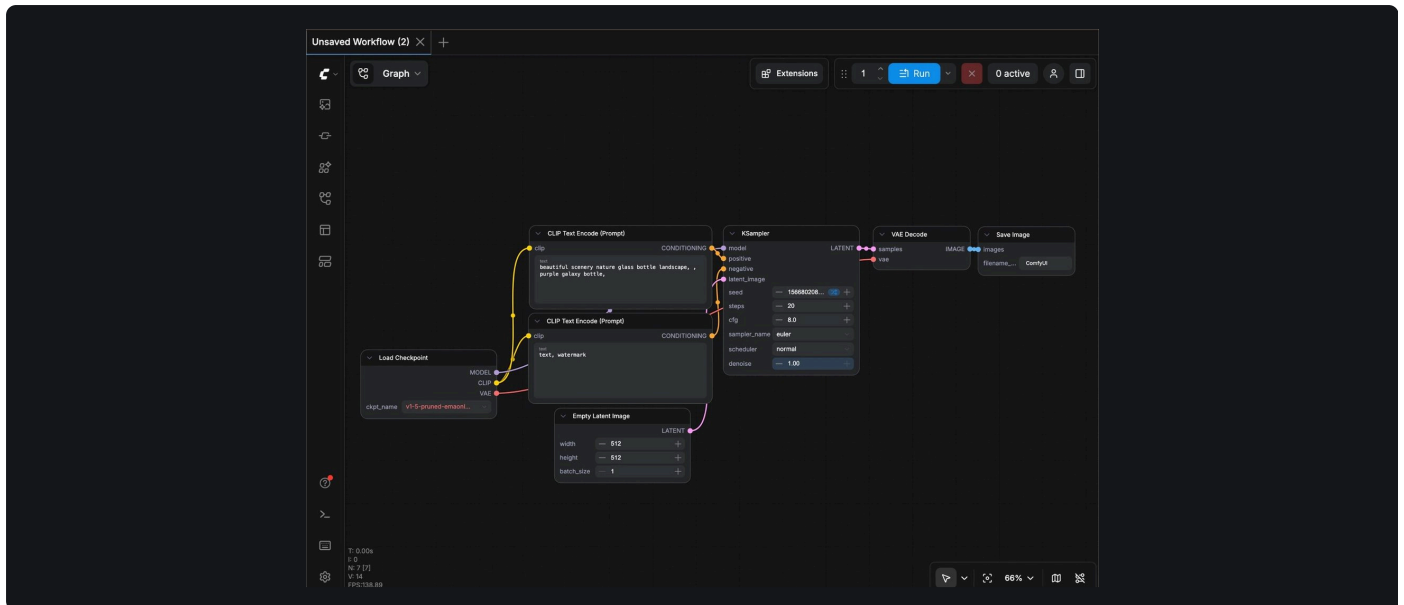


Search-to-place. Typing filters the whole operator library inline on the canvas, and the result drops where you were looking.

STEAL THIS FOR ESKIMO

This is the fix for diagnosis #1. Eskimo's six socket rows — roughly 60% of every card's height — could collapse into two edge strips a few pixels tall. The type is the colour; the label appears on hover or on the detail pane. Rectangular ports are also a *larger* hit target than the current 9px circles despite taking less space, which incidentally helps the drag-connect problem.

The most-used node editor of the last few years, and a good model for a modern web-native one. Its nodes are *forms*: they carry real controls, not just wires.



ComfyUI. Typed sockets carry the type name in caps as the label — MODEL, CLIP, VAE, LATENT, IMAGE, CONDITIONING — with the wire in the matching colour. Inline widgets (text areas, number steppers with $\pm$, dropdowns) live in the node body. Each node has a collapse chevron in its title bar.

THE DESIGN SYSTEM

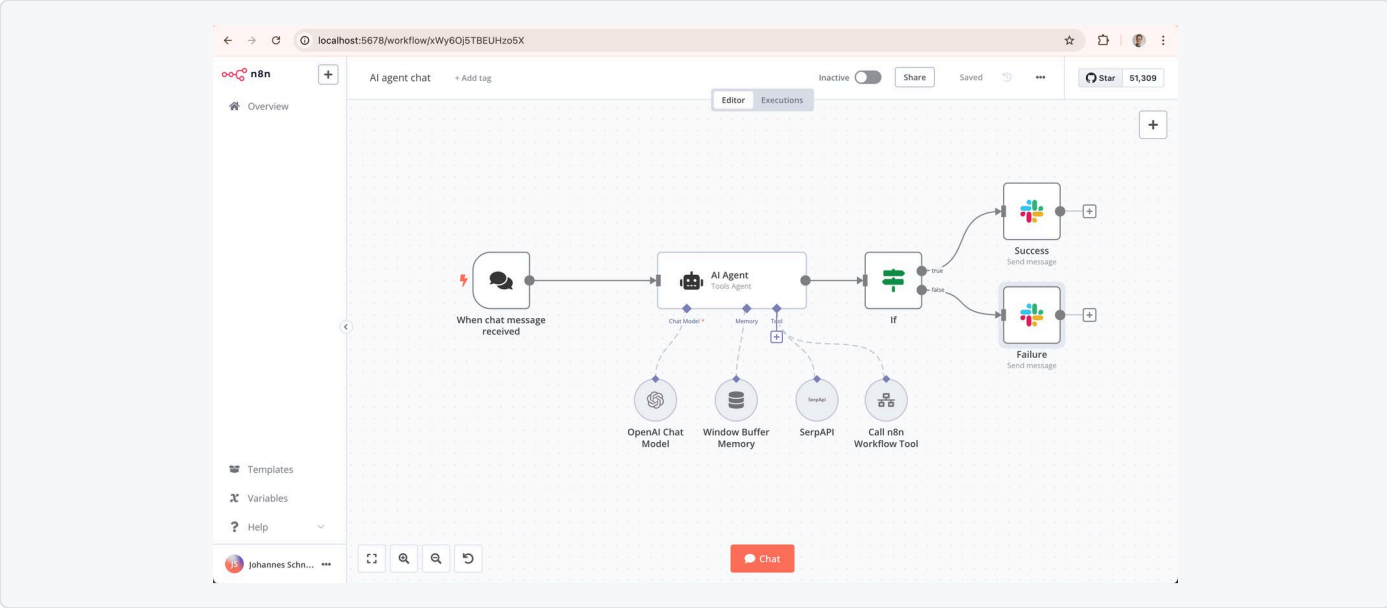
- **Sockets are named by type, and the name is on the node**, right-aligned for outputs and left-aligned for inputs, so the reading order matches the flow.
- **Widgets in the body.** Editing the common parameters never requires opening a side panel.
- **Restrained node chrome, saturated wires.** The nodes are near-black; all the colour is in the connections.

STEAL THIS FOR ESKIMO

Inline controls. Eskimo's variant picker already lives on the node — that instinct was right. The natural extension is the cue point: a draggable in/out marker directly on the card's waveform, rather than a number in a pane.

Let the wires carry the saturation and keep the cards quiet. This is the opposite of Eskimo's current split, where the cards carry a colour swatch and the wires are black.

The single most relevant reference here, because it is the one modern, polished, **light-themed** node editor in the study — which is the mode Eskimo actually ships in.



n8n. Nodes are small rounded squares carrying an icon. The name sits *outside and below* the node. Outputs are labelled only where the label means something ("true" / "false"). Sub-connections use dashed wires and circular nodes to mark a different class of relationship. Unconnected outputs carry a + button that adds the next step.

UX FEATURES WORTH NOTING

FEATURE	WHY IT WORKS
Label outside the node	The node box stays small and uniform; long names cost nothing because they overflow into empty canvas.
The + on an open output	Turns an empty socket into an invitation. Building a chain never requires finding the node palette.
Dashed vs solid wires	Two connection classes, distinguished without spending a colour.
Selective output labels	Only branching nodes label their outputs. Everything else stays clean.
Light canvas, dotted grid	Proves a light node editor can look modern — it just needs real spacing and a value gap between card and canvas.

STEAL THIS FOR ESKIMO

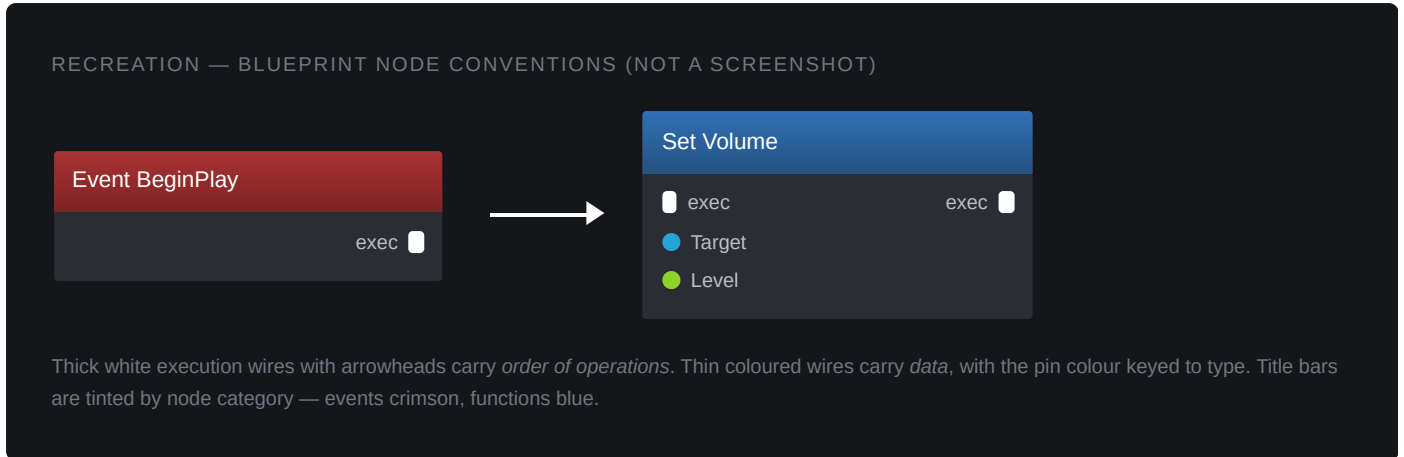
- Move the song title outside the card.** Eskimo's cards are sized by their text. Put the title and artist below the card and the card itself can shrink to artwork plus waveform — a large density win that costs nothing.
- Add the + affordance to unwired outputs.** Eskimo already knows which outputs are open and what could legally connect there.
- Label outputs only when they branch.** Directly retires the ~100 redundant "None" labels.

11 Unreal Engine — Blueprints

Epic Games · visual scripting

COLOURFUL AND INTUITIVE

Blueprints solve a problem Eskimo has exactly: **two different kinds of connection that must never be confused**. Unreal's answer is to give them different physical weight, not just different colour.



UX FEATURES WORTH NOTING

- **Execution wires are visually heavier than data wires** and carry direction arrows. You can read the control flow of a huge graph at a glance while ignoring the data plumbing.
- **Comment boxes** are coloured, titled, and dragging one moves everything inside it — grouping without a formal container type.
- **Wire animation during debugging** shows execution travelling along the path in real time.

STEAL THIS FOR ESKIMO

Two weights for two meanings. Eskimo already has this distinction and doesn't draw it: the *committed* playback path (now frozen in `session.committedHop`) versus every other *possible* path. Committed should be heavy with direction; possible should be a hairline. That one change would make "where is this set going" readable from across the room — which for a live DJ tool is the entire job.

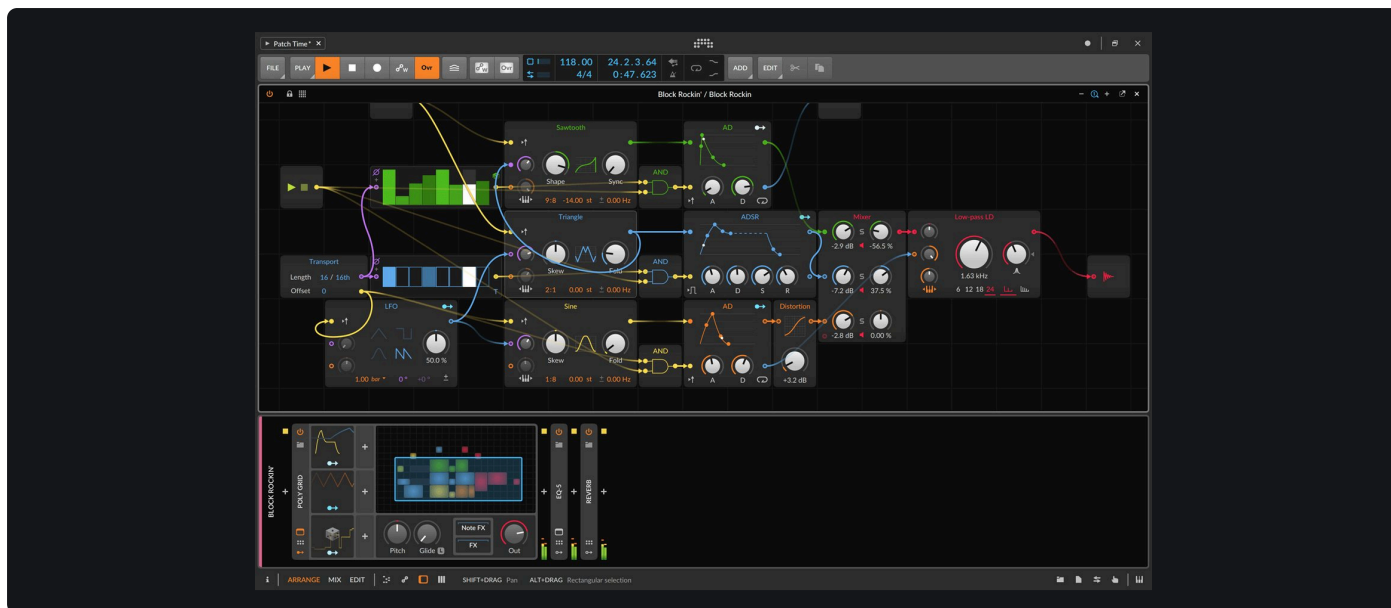
The debug wire animation is the model for live playback — and Eskimo already has an animated active-path edge, so this instinct is confirmed by a tool that ships it to millions of users.

12 Bitwig Studio — The Grid

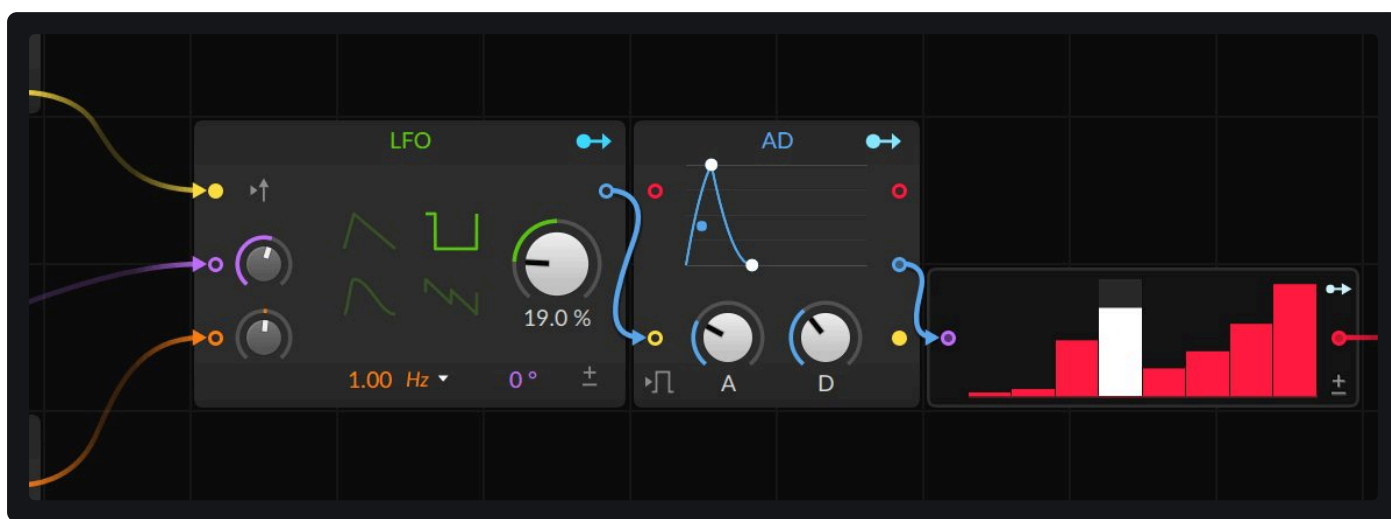
Modular audio inside a DAW

COLOURFUL AND INTUITIVE

The closest peer in the entire study: a node editor for *audio*, built for people who are also DJs and producers, and designed to be legible while something is playing.



The Grid. Wire colour encodes signal type. Nodes carry real inline controls — knobs, editable envelope curves, waveform selectors — and the node title is tinted to match its module category.



Signal types as a colour system. The palette is the specification — which outputs may meet which inputs is answered by colour before you ever attempt the connection.

STEAL THIS FOR ESKIMO

The whole signal-type colour model transfers directly. Eskimo's four connection types are its signal types. Bitwig demonstrates that in an audio tool this reads as professional rather than childish, provided the palette is disciplined — saturated wires, desaturated everything else.

Inline controls read as an instrument. Bitwig's nodes look like gear. Eskimo's look like a spreadsheet row. A waveform, a cue marker and a small level readout would close most of that gap.

What they all agree on

Individual tricks are interesting; unanimity is actionable. Five patterns appear in essentially every tool studied — and Eskimo does none of them.

PATTERN	WHO DOES IT	ESKIMO TODAY
Connections are colour-coded by type	Blender, ComfyUI, Bitwig, Unreal, cables.gl, TouchDesigner	All wires identical black
A per-node density control	Blender (collapse), Nuke (stamps off), Houdini (badges scale), ComfyUI (chevron)	None — always maximum detail
The node displays its own content	TouchDesigner, Nuke, Substance, ComfyUI, Bitwig	Album art only — decorative, not informative
Labels only where they carry information	n8n, cables.gl, Nuke, Houdini	Six labels per card regardless of state
Node size or shape varies with role	Blender, Unreal, ComfyUI, n8n, Bitwig	One fixed card size for everything

The one genuine disagreement

Canvas value. Eight of the ten use a dark canvas; n8n is the notable exception and it looks excellent. So **light is a legitimate choice** — but it is the harder one, because a light canvas gives you a much smaller value range to separate card from background. n8n buys that separation with generous whitespace, a crisp card border radius, and by keeping nodes small. Eskimo currently pays for a light canvas without buying any of the three.

The short version

Eskimo doesn't look unprofessional because it's under-designed. It looks unprofessional because **it spends its space on labels and its colour on decoration**, when every tool that reads as professional does the reverse: space goes to content, colour goes to meaning.

Recommendations for Eskimo

Ordered by impact per unit of work. The first four are the ones that would change how the app feels; the rest are refinement.

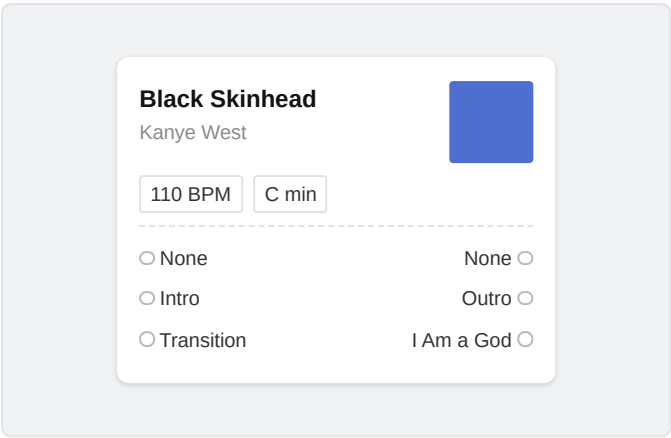
- 1 **Collapse the socket matrix into edge strips.**
cables.gl · Two thin coloured strips instead of six labelled rows. Recovers roughly 60% of card height, removes ~300 rows of text from the canvas, and gives a bigger hit target than the current circles. Highest impact by a wide margin.
- 2 **Colour wires by connection type; two weights for committed vs possible.**
Bitwig · Unreal · Blender · Four hues for None / Intro / Outro / Transition. Heavy stroke with direction for the committed path, hairline for everything else. Pure rendering change — no data model work, since committedHop already exists.
- 3 **Put the waveform on the card.**
TouchDesigner · Nuke · With the splice point marked. The envelope data and the renderer already exist for the transition preview. Make it disable-able on large graphs, as Nuke does.
- 4 **Move title and artist outside the card.**
n8n · Lets the card shrink to art plus waveform and stops long titles from driving layout.
- 5 **Thin the wires and improve routing.**
Audulus · 1–1.5px with a soft outer stroke so crossings stay readable, and enough curvature that wires stop cutting through unrelated cards.
- 6 **Add a badge row.**
Houdini · Has audio, verified splice, local-only, contributed — four marks replacing four different text treatments.
- 7 **Collapse-to-pill per node.**
Blender · The user's own density control, and the natural home for a zoom-based level-of-detail rule later.
- 8 **A + affordance on unwired outputs.**
n8n · Turns an empty socket into the primary way to build a set.
- 9 **Escalate socket targets on hover.**
Houdini's node ring · Fixes small-target hit testing at low zoom without enlarging the resting state.
- 10 **A permanent category strip for adding nodes.**
Audulus · Makes the app's vocabulary visible instead of hiding it behind a right-click.

Sequencing note. Items 1–4 are mutually reinforcing and should be designed together — shrinking the card, moving the label out, and adding the waveform are really one redesign of the node, not four. Item 2 is independent and could ship on its own tomorrow.

Proposed node anatomy

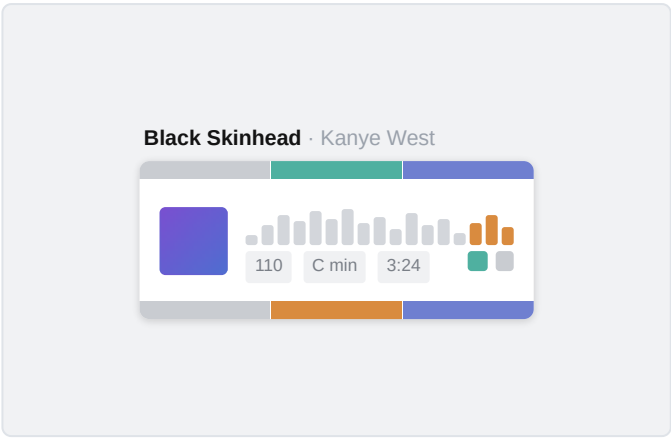
A sketch, not a spec — recommendations 1 through 6 applied to a single card so the combined effect is visible. Both are drawn at the same scale.

TODAY



Six socket rows, ~60% of the height. Title drives the width. The only colour is decorative. Nothing indicates what the audio actually is.

PROPOSED



Ports become edge strips. Title moves outside. The waveform shows the outro region in its own type colour. Badges replace text. Roughly 45% shorter.

None Intro Outro Transition — the same four colours on the wires

What this does not solve

Edge strips lose the ability to label a specific target song on the card ("Transition → I Am a God"). That information has to move somewhere — hover, the detail pane, or a label on the wire itself. Wire labels are what Blender and Unreal would do, but they add their own clutter at scale. **This is a real trade-off and worth deciding deliberately rather than by default.**

The current build already has the hard parts: a committed-hop model, an envelope renderer, dominant-colour extraction, dark theme tokens, and a working drag-connect. Every recommendation here is presentation sitting on top of those.

Sources

Screenshots are from official documentation and product pages, reproduced here for design study. Three tools are illustrated with labelled recreations rather than screenshots.

SCREENSHOTS

Blender	— docs.blender.org/manual · node parts, socket types, editing, shader editor
TouchDesigner	— docs.derivative.ca · Operator, TOP, viewers
ComfyUI	— docs.comfy.org · interface overview
n8n	— github.com/n8n-io/n8n · README workflow screenshot
Bitwig Studio	— bitwig.com/the-grid · product pages
cables.gl	— cables.gl · interface captures
Audulus	— audulus.com · editor and wire treatment
Eskimo Studio	— captured from the current build at 2× on the example graph

RECREATIONS (DRAWN FOR THIS DOCUMENT)

Nuke	— node classes and postage stamps, per Foundry documentation and support notes on disabling or freezing thumbnails in large scripts
Houdini	— flags, badges and the hover ring, per SideFX network editor documentation
Unreal Blueprints	— execution versus data wires and pin typing, per long-standing Blueprint conventions

WRITTEN REFERENCES

SideFX	— Houdini network editor: nodes, flags and badges
Foundry Learn	— Working with Nodes; PostageStamp node reference
Foundry Support Q100036	— disabling thumbnails or setting static frames for script performance
Blender Manual	— Interface › Controls › Nodes › Parts of a node

Compiled for Eskimo Studio, September 2026. All product names and screenshots remain the property of their respective owners and appear here solely as design reference.